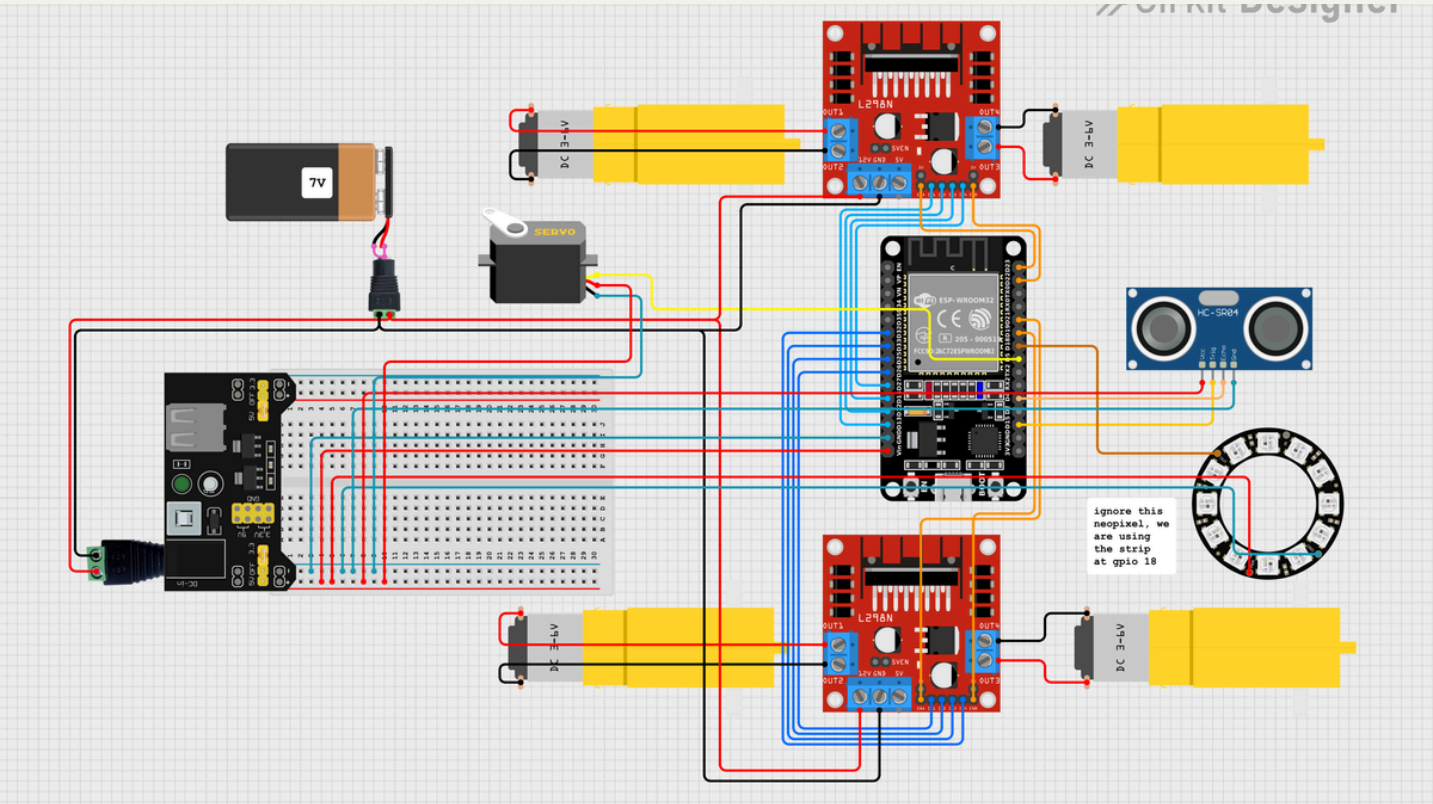
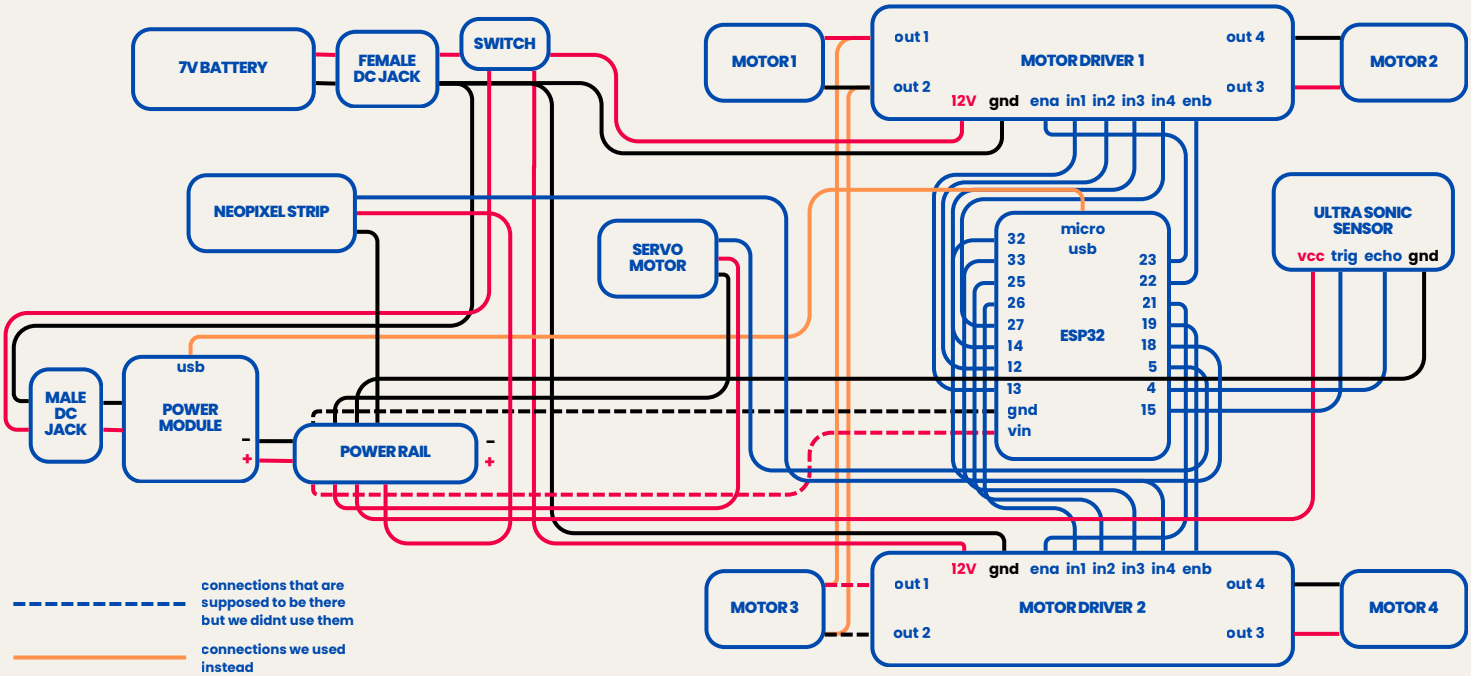
Electronics Details

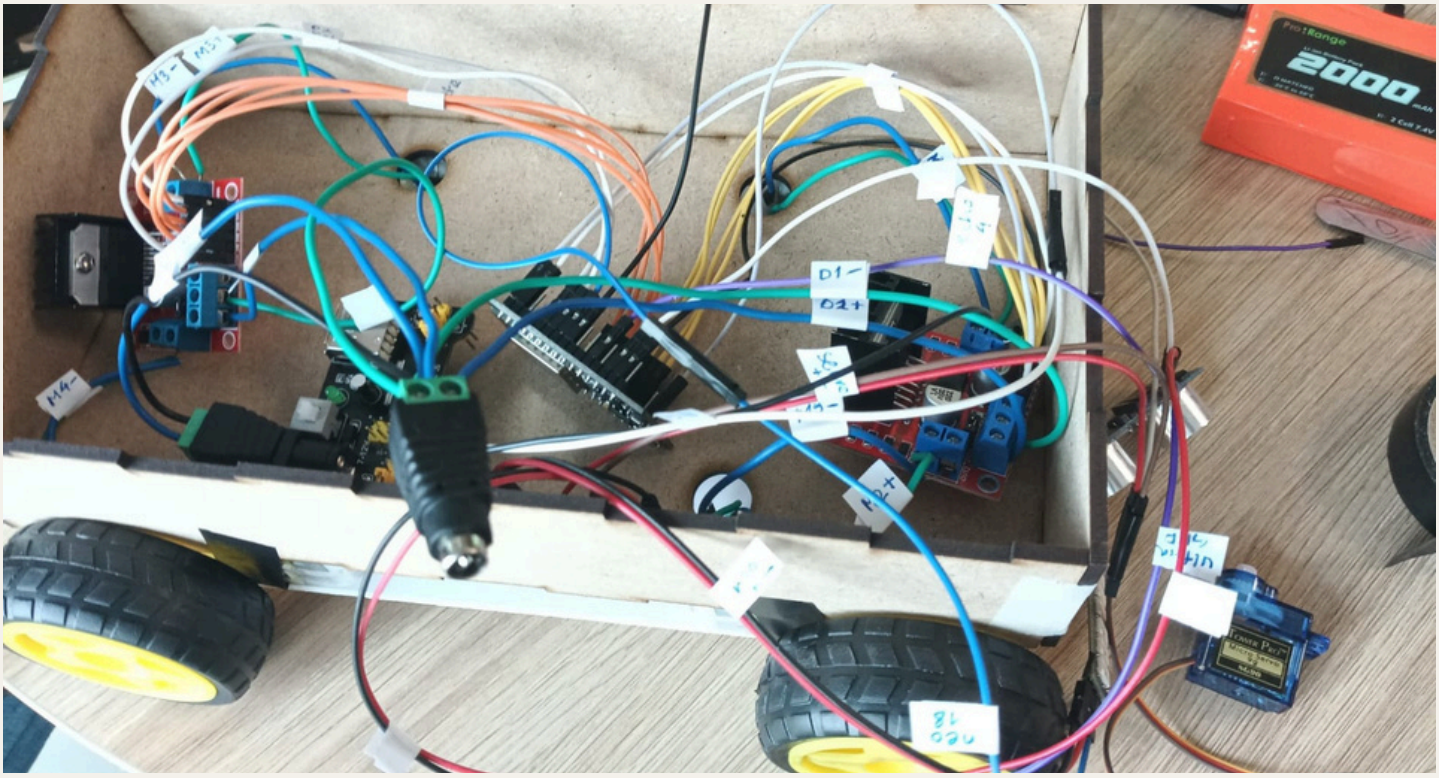
CIRCUIT DIAGRAM



BLOCK DIAGRAM



ELECTRONIC DETAILS

COMPONENTS	QUANTITY	PURPOSE	JUSTIFICATION
ESP32 Microcontroller	01	Processes all inputs, runs the code, and controls every output	Chosen for its built-in Bluetooth Low Energy (BLE), dual-core processor, and large number of GPIO pins with PWM support. Allowed wireless control via the app without any extra modules
Power Supply Module	01	Regulates the battery voltage down to a stable 5V for components that can't run directly off the battery	The battery outputs between 3V–7.4V which is too unstable and too high for the ESP32, ultrasonic sensor, servo, and NeoPixel strip. The module provides clean regulated 5V to all of these
7.4V Battery	01	Main power source for the entire car	Acts as an external power supply for the car and powers the motor drivers directly and feeds the power supply module for the rest of the circuit
L298N Motor Driver	02	Acts as an intermediary between the ESP32 and the DC motors	Two drivers were needed to independently control all four wheels. Each driver handles two motors, allowing individual speed and direction control which is essential for turning
BO DC Gear Motor	04	Drives and rotates each wheel independently	BO gear motors are lightweight and compact, making them well suited for a small chassis. They support variable speed control via PWM signals and provide enough torque for the cars weight
Jumper Wires & Copper Wire	a lot	Connect all components across the circuit	Jumper wires used for standard connections, copper wire used where a custom-length were required. All wires were individually labelled to keep the circuit traceable
Male & Female DC Jack	01 each	Connects the battery and the power module supply to the rest of the circuit	The female dc jack allows multiple wires to connect to the battery and the male one allows the power module to connect to the battery
HC-SR04 Ultrasonic Sensor	01	Detects obstacles in front of the car by measuring distance using sound pulses	Used for detecting and measuring the distance between the car and the obstacle at a given threshold
Physical Switch	01	Breaks the main power line to turn the car on and off	Outside the chassis so the car can be powered off without opening the enclosure or disconnecting the battery manually also prevents unnecessary battery drain between uses

Servo Motor	01	Rotates the ultrasonic sensor so it can scan a wider angle rather than only detecting obstacles directly in front	A single fixed-facing sensor has a narrow detection cone. Mounting it on a servo lets the sensor sweep across a range, giving the car better spatial awareness in auto mode
Breadboard Power Rail	01	Distributes GND and 5V from the power supply module to multiple components at once	A full breadboard wouldn't fit inside the enclosure. Using just the power rail strips gave a clean shared bus for all the 5V and GND connections without taking up extra space
USB to ESP32 Cable	01	Used to deliver power from the power supply module directly to the ESP32	The ESP32 was not powering on reliably when connected through the VIN pin alone. Routing power through the USB port gave a more stable and direct power path. The cable was repurposed from its original use (laptop connection) and wired into the power supply output
NeoPixel LED Strip	01	Provides a visual indicator of the car's current state	Gives immediate visual feedback without needing to check the app. Each colour maps to a specific state (e.g. moving forward, turning, obstacle detected, idle) making it easy to read the car's behaviour at a glance
Electrical Tape	01	Insulates exposed wire joints and secures loose connections	Prevented short circuits in areas where bare wire ends were close together inside the compact enclosure